MEI FM Numerical methods: Numerical integration

Section 2: Simpson's rule

Notes and Examples

These notes contain the following subsections:

Simpson's rule

Simpson's rule in terms of the midpoint rule and the trapezium rule

Simpson's rule

In the previous section, you met the midpoint rule and trapezium rule which can be used to calculate numerical estimates for a definite integral.

Another rule that can be used is Simpson's rule, which usually gives a more accurate estimate for the integral.

In Simpson's rule, an even number of strips must be used. You will see why this is the case shortly.

The Simpson's rule approximation S_{2n} to the integral $\int_a^b \mathrm{f}(x)\,\mathrm{d}x$ is found as follows.

Divide the interval from a to b into $2n$ strips, each of width $h = \dfrac{b-a}{2n}$. As usual let f_0 be the value of the function at a, f_1 the value at the right hand end of the first strip and so on, so that f_{2n} is the value of the function at b. Then:

$$S_{2n} = \frac{h}{3}\left[\mathrm{f}_0 + \mathrm{f}_{2n} + 4\left(\mathrm{f}_1 + \mathrm{f}_3 + \mathrm{f}_5 + \ldots + \mathrm{f}_{2n-1}\right) + 2\left(\mathrm{f}_2 + \mathrm{f}_4 + \mathrm{f}_6 + \ldots + \mathrm{f}_{2n-2}\right)\right]$$

How this formula is established is described below.

Simpson's rule relies on the fact that, given three points in the plane, there is a unique quadratic or straight line, which passes through them. Simpson's rule works by approximating the graph of any function as a series of quadratics or straight lines, connected end to end. Each quadratic involves three points and therefore two strips, which explains why there must be an even number of strips.

The diagram below explains this for the S_2 estimate to $\int_a^b \mathrm{f}(x)\,\mathrm{d}x$, which is the shaded area shown.

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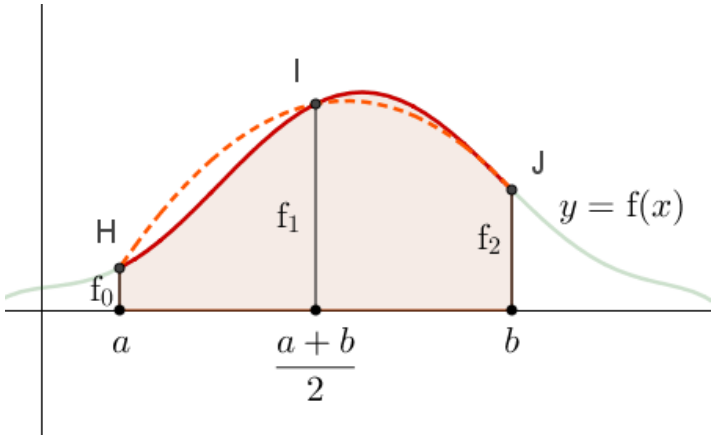
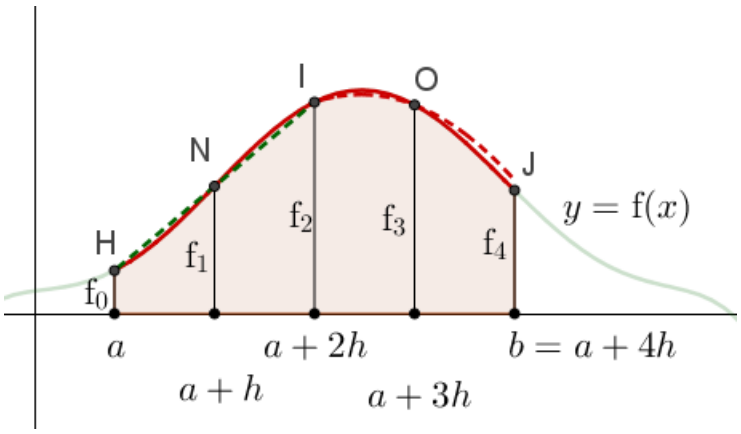


The orange dotted line is the graph of a quadratic function which fits through the points H, I and J. You will learn more about how polynomials can be used to approximate other functions in a later topic. The S_2 estimate to $\int_a^b f(x)\,dx$ is the area under this quadratic which can be calculated by direct integration. It turns out the area is $\frac{h}{3}(f_0 + 4f_1 + f_2)$ where $h = \frac{b-a}{2}$ (you can research this if you want to know more about why this is so).

The S_4 estimate to $\int_a^b f(x)\,dx$ uses two quadratics.

In the diagram below, where $h = \frac{b-a}{4}$, there is a quadratic (shown as a dotted green line) passing through H, N and I and another quadratic (shown as a dotted brown line) passing through I, O and J.



$$S_4 = (\text{area under quadratic through H,N,I}) + (\text{area under quadratic through I,O,J})$$

$$= \frac{h}{3}(f_0 + 4f_1 + f_2) + \frac{h}{3}(f_2 + 4f_3 + f_4)$$

$$= \frac{h}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + f_4)$$

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In general, dividing the region from $x = a$ to $x = b$, into an even number, $2n$, of strips, of width h, Simpson's rule gives;

$$\int_a^b f(x)\,dx = \frac{h}{3}\left[f_0 + 4(f_1 + f_3 + f_5 + \ldots + f_{2n-1}) + 2(f_2 + f_4 + \ldots + f_{2n-2}) + f_{2n}\right]$$

You may find the following tables useful as a guide for setting out your own tables or spreadsheets when using Simpson's rule to get S_2, S_4, S_6 and S_8.

2 strips

x	f(x)	Factor	Product
x_0	f_0	1	f_0
$x_1=x_0+h$	f_1	4	$4f_1$
$x_2=x_1+h$	f_2	1	f_2

Sum = $f_0+4f_1+f_2$

$$\int_{x_0}^{x_2} f(x)dx \approx \frac{h}{3}(f_0 + 4f_1 + f_2)$$

4 strips

x	f(x)	Factor	Product
x_0	f_0	1	f_0
$x_1=x_0+h$	f_1	4	$4f_1$
$x_2=x_1+h$	f_2	2	$2f_2$
$x_3=x_2+h$	f_3	4	$4f_3$
$x_4=x_3+h$	f_4	1	f_4

Sum = $f_0+4f_1+2f_2+4f_3+f_4$

$$\int_{x_0}^{x_4} f(x)dx \approx \frac{h}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + f_4)$$

6 Strips

x	f(x)	Factor	Product
x_0	f_0	1	f_0
$x_1=x_0+h$	f_1	4	$4f_1$
$x_2=x_1+h$	f_2	2	$2f_2$
$x_3=x_2+h$	f_3	4	$4f_3$
$x_4=x_3+h$	f_4	2	$2f_4$
$x_5=x_4+h$	f_5	4	$4f_5$
$x_6=x_5+h$	f_6	1	f_6

Sum= $f_0+4f_1+2f_2+4f_3+2f_4+4f_5+f_6$

$$\int_{x_0}^{x_6} f(x)dx \approx \frac{h}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 + 4f_5 + f_6)$$

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8 Strips

x	f(x)	Factor	Product
x_0	f_0	1	f_0
$x_1=x_0+h$	f_1	4	$4f_1$
$x_2=x_1+h$	f_2	2	$2f_2$
$x_3=x_2+h$	f_3	4	$4f_3$
$x_4=x_3+h$	f_4	2	$2f_4$
$x_5=x_4+h$	f_5	4	$4f_5$
$x_6=x_5+h$	f_6	2	$2f_6$
$x_7=x_6+h$	f_7	4	$4f_7$
$x_8=x_7+h$	f_8	1	f_8

Sum $= f_0+4f_1+2f_2+4f_3+2f_4+4f_5+2f_6+4f_7+f_8$

$$\int_{x_0}^{x_6} f(x)dx \approx \frac{h}{3}(f_0+4f_1+2f_2+4f_3+2f_4+4f_5+2f_6+4f_7+f_8)$$

Simpson's rule in terms of the midpoint rule and the trapezium rule

Simpson's rule can be expressed in terms of the formulae for the midpoint rule and the trapezium rule.

If you are calculating S_8, the values of the function that you need are $f_0,\ f_1,\ f_2,\ldots,f_8$.

$$S_8 = \frac{1}{3}\times\frac{b-a}{8}\left[f_0+4(f_1+f_3+f_5+f_7)+2(f_2+f_4+f_6)+f_8\right]$$

If you use the midpoint rule with 4 strips to approximate the same integral you use the values, $f_1, f_3, \ldots, f_7$.

$$M_4 = \frac{b-a}{4}\left(f_1+f_3+f_5+f_7\right)$$

Using the trapezium rule with 4 strips requires the values $f_0, f_2, \ldots, f_8$.

$$T_4 = \frac{1}{2}\times\frac{b-a}{4}\left[f_0+2(f_2+f_4+f_6)+f_8\right]$$

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You can see that $T_4 + 2M_4 = \frac{b-a}{8}\left[\mathrm{f}_0 + 2(\mathrm{f}_2 + \mathrm{f}_4 + \mathrm{f}_6) + \mathrm{f}_8 + 4\left(\mathrm{f}_1 + \mathrm{f}_3 + \mathrm{f}_5 + \mathrm{f}_7\right)\right]$, and so

$$\frac{T_4 + 2M_4}{3} = \frac{1}{3} \times \frac{b-a}{8}\left[\mathrm{f}_0 + 4\left(\mathrm{f}_1 + \mathrm{f}_3 + \mathrm{f}_5 + \mathrm{f}_7\right) + 2(\mathrm{f}_2 + \mathrm{f}_4 + \mathrm{f}_6) + \mathrm{f}_8\right] = S_8.$$

Generalising this, there is a relationship between S_{2n} (Simpson's rule with $2n$ strips), M_n (the midpoint rule with n strips) and T_n (the trapezium rule with n strips).

$$S_{2n} = \frac{T_n + 2M_n}{3}.$$

In addition, you can probably spot that there is a relationship between M_n and T_n.

$$\begin{aligned} T_4 + M_4 &= \frac{1}{2} \times \frac{b-a}{4}\left[\mathrm{f}_0 + 2(\mathrm{f}_2 + \mathrm{f}_4 + \mathrm{f}_6) + \mathrm{f}_8 + 2\left(\mathrm{f}_1 + \mathrm{f}_3 + \mathrm{f}_5 + \mathrm{f}_7\right)\right] \\ &= 2 \times \frac{1}{2} \times \frac{b-a}{8}\left[\mathrm{f}_0 + 2(\mathrm{f}_1 + \mathrm{f}_2 + \mathrm{f}_3 + \mathrm{f}_4 + \mathrm{f}_5 + \mathrm{f}_6 + \mathrm{f}_7) + \mathrm{f}_8\right] \\ &= 2T_8 \end{aligned}$$

So $T_8 = \frac{T_4 + M_4}{2}$

which can be generalised to $T_{2n} = \frac{T_n + M_n}{2}$.

You can then eliminate M_n from $S_{2n} = \frac{T_n + 2M_n}{3}$ and $T_{2n} = \frac{T_n + M_n}{2}$ to give

$$S_{2n} = \frac{4T_{2n} - T_n}{3}$$

These relationships are more than just interesting identities. They can save you a lot of time.

Example 1

Calculate S_2, S_4 and S_8 for the integral $\int_1^2 \sqrt{\sin x}\,\mathrm{d}x$

(a) directly from function values

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(b) indirectly from midpoint and trapezium rule estimates using the formula

$$S_{2n} = \frac{T_n + 2M_n}{3}.$$

Solution

The calculations are shown in the spreadsheet view below. The formulae used are also shown.

(a)

	A	B	C	D	E
1	x	$\sqrt{\sin x}$		S_2	0.977645832
2	1	0.917317276		S_4	0.977738156
3	1.125	0.949877673		S_8	0.977745267
4	1.25	0.974158416			
5	1.375	0.990400453			
6	1.5	0.998746708			
7	1.625	0.9992654			
8	1.75	0.991960658			
9	1.875	0.976773147			
10	2	0.953570882			

	A	B	C	D	E
1	x	$\sqrt{\sin x}$		S_2	=(0.5/3)*(B2+4*B6+B10)
2	1	=SQRT(SIN(A2))		S_4	=(0.25/3)*(B2+4*(B4+B8)+2*B6+B10)
3	1.125	=SQRT(SIN(A3))		S_8	=(0.125/3)*(B2+ 4*(B3+B5+B7+B9)+2*(B4+B6+B8)+B10)
4	1.25	=SQRT(SIN(A4))			
5	1.375	=SQRT(SIN(A5))			
6	1.5	=SQRT(SIN(A6))			
7	1.625	=SQRT(SIN(A7))			
8	1.75	=SQRT(SIN(A8))			
9	1.875	=SQRT(SIN(A9))			
10	2	=SQRT(SIN(A10))			

(b)

	A	B	C	D	E
1	x	$\sqrt{\sin x}$		M_1	0.998746708
2	1	0.917317276		M_2	0.983059537
3	1.125	0.949877673		M_4	0.979079169
4	1.25	0.974158416			
5	1.375	0.990400453		T_1	0.935444079
6	1.5	0.998746708		T_2	0.967095393
7	1.625	0.9992654		T_4	0.975077465
8	1.75	0.991960658			
9	1.875	0.976773147		S_2	0.977645832
10	2	0.953570882		S_4	0.977738156
11				S_8	0.977745267

	A	B	C	D	E
1	x	$\sqrt{\sin x}$		M_1	=1*B6
2	1	=SQRT(SIN(A2))		M_2	=0.5*(B4+B8)
3	1.125	=SQRT(SIN(A3))		M_4	=0.25*(B3+B5+B7+B9)
4	1.25	=SQRT(SIN(A4))			
5	1.375	=SQRT(SIN(A5))		T_1	=0.5*(B2+B10)
6	1.5	=SQRT(SIN(A6))		T_2	=(E1+E5)/2
7	1.625	=SQRT(SIN(A7))		T_4	=(E2+E6)/2
8	1.75	=SQRT(SIN(A8))			
9	1.875	=SQRT(SIN(A9))		S_2	=(2*E1+E5)/3
10	2	=SQRT(SIN(A10)		S_4	=(2*E2+E6)/3
11				S_8	=(2*E3+E7)/3

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